MILITARY SPECIFICATION

ANTENNA ASSEMBLY, BUOYANT CABLE, AS-2629B/BRR

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabrication, performance, test, and acceptance requirements for the Antenna Assembly, Buoyant Cable, AS-2629B/BRR, hereinafter referred to as the antenna assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triple-wall
PPP-S-760	Strapping, Nonmetallic (And Connectors)
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-C-12000	Cable, Cord, And Wire, Electric; Packaging Of
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For

NAVAL UNDERWATER SYSTEMS CENTER (NUSC), New London Laboratory (NLL)

NUSC-C-342/4141-279/1-10	Cable, Radio Frequency, Type I, NUSC-C-279/1-10
NUSC-3-342/4140-279/2-12	Cable, Radio Frequency, Coaxial, Type II, NUSC-C-279/2-12

STANDARD

MILITARY

MIL-STD-109	Quality Assurance Terms And Definitions

2.1.2 Other Government drawings (DWGs) The following other Government DWGs form a part of this specification to the extent specified herein.

NUSC NLL

NUSC/NLL 02202001	Tip, Cable, Termination, (Grounded)
NUSC/NLL C 02377003	Cover, Test/Shipping
NUSC/NLL D 02379001	Connector, Plug, Electrical (Pin Contact)
NUSC/NLL D 02380001	Housing Assembly
NUSC/NLL D 02380005	Liner
NUSC/NLL C 02381001	Test Plug

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5985

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NUSC/NLL A 02710001	Amplifier, R.F., Assembly Procedure
NUSC/NLL D 02382001	Connector, Plug, Electrical (Socket Contact)
NUSC/NLL D 02383001	Adapter, Connector (Pin Contact)
NUSC/NLL D 02384001	Adapter, Connector (Socket Contact)
NUSC/NLL D 03055001	Amplifier, R.F., Connector Interface and Chassis Arrangement
NUSC/NLL D 03055004	Amplifier, R.F., AM-7041/BRR Schematic Diagram
NUSC/NLL D 03056001	Antenna Assembly, Buoyant Cable, AS-2629/BRR
NUSC/NLL D 03056002	R.F. Cable Assembly, CG-3816/BRR
NUSC/NLL D 03056003	Antenna Element Assembly AS-3376/BRR

(Copies of specifications, standards, and DWGs required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The antenna assembly shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Description. The antenna assembly shall consist of the components specified in FIGURE 1.

3.2 Characteristics. Characteristics shall be as specified in 3.2.1 through 3.2.3.5.

3.2.1 Performance. The antenna assembly shall receive very low frequency (VLF) and low frequency (LF) communication and navigation signals on the braid of the radio frequency (RF) Cable Assembly, CG-3816/BRR and medium frequency (MF) and high frequency (HF) communications up to 30 megahertz (MHz) on the Antenna Element Assembly, AS-3376/BRR. The antenna assembly shall provide amplification which increases with the frequency of the MF and HF signals as specified in 3.2.1.2. The antenna assembly shall interface electrically with Antenna Coupler, CU-2065/BRR or Antenna Coupler, CU-1441A/BRR, and shall be mechanically compatible with the Antenna Transfer Assemblies, AN/BRA-18( ) and AN/BRA-24( ).

3.2.1.1 Burn-in. The amplifier shall operate satisfactorily after being subjected to a 24-hour burn-in test as specified in 4.5.1.

3.2.1.2 Voltage gain. When tested in accordance with 4.5.6, the RF amplifier shall exhibit a measured voltage gain within the limits specified in TABLE I.

TABLE I. Gain requirements.

Frequency MHz	Voltage gain, 50-ohm load dB 1/ ±4 dB
0.5	17
1.0	21
2.0	22
5.0	25
10.0	29
20.0	34
30.0	35

1/ Decibels

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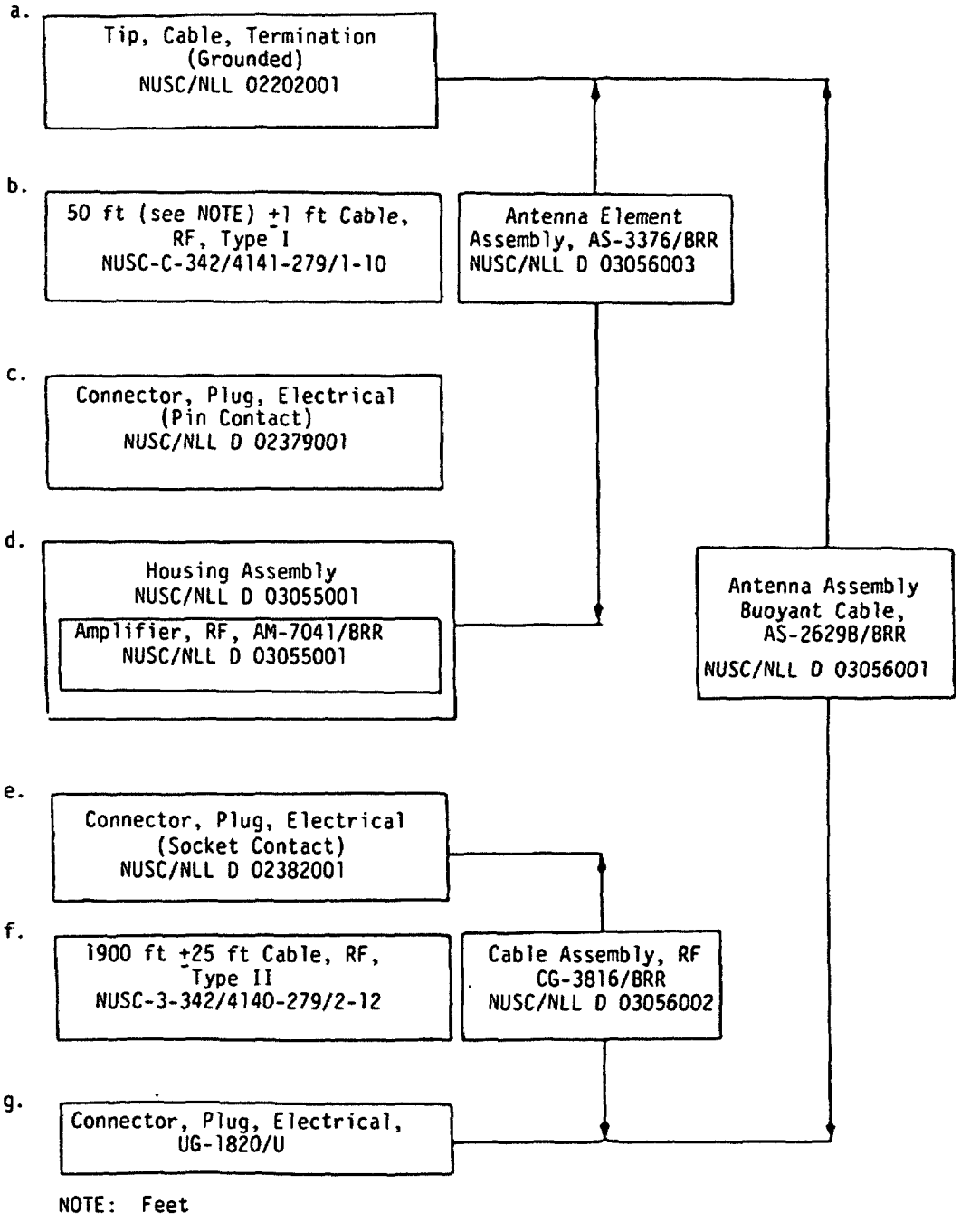


FIGURE 1. Components of antenna assembly.

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3.2.1.3 Stability. The RF amplifier shall exhibit no discrete output signals when tested in accordance with 4.5.7.

3.2.1.4 Noise figure. When tested in accordance with 4.5.8, the RF amplifier shall have a noise figure no greater than that specified in TABLE II.

TABLE II. Noise figure requirements.

Frequency MHz	Noise figure at room temperature (20°C 1/) dB
2	7.0
10	6.0
30	5.0

1/ Celsius

3.2.1.5 Gain compression. When tested in accordance with 4.5.9, the RF amplifier shall exhibit an output level of 0 decibels referred to 1 milliwatt (dBm), minimum, for a gain compression of 1 dB.

3.2.2 Physical characteristics. Physical characteristics shall be as specified in 3.2.2.1 and 3.2.2.2.

3.2.2.1 Dimensions. Unless otherwise specified in the contract (see 6.2.1), the dimensions of the antenna assembly shall be in accordance with NUSC/NLL D 03056001 and this specification.

3.2.2.2 Health and safety. Health and safety considerations require avoidance of toxicological substances such as encapsulants, adhesives, or preservatives which could contaminate the closed environment of a submarine. The materials shown on the DWGs are acceptable. Substitutes shall be approved by the Government.

3.2.3 Environmental conditions. Environmental conditions shall be as specified in 3.2.3.1 through 3.2.3.5.

3.2.3.1 Temperature. The temperature range of the antenna assembly shall be Range 3 of MIL-E-16400. The antenna shall suffer no damage when stored on a reel with a 15-inch (in.) hub diameter at any of the nonoperating temperatures of Range 3.

3.2.3.2 Tensile strength. When tested in accordance with 4.5.2, the connector plug, electrical (pin contact); RF amplifier, and connector, plug, electrical (socket contact) with suitable test fixtures attached shall withstand static tensile loads along the axis with no evidence of cracking or other damage. The maximum elongation of the molded section including the separation of connector (metal section) from the antenna assembly shall not exceed 2 percent when subjected to tensile loads specified in 4.5.2.

3.2.3.3 Flexibility. When tested in accordance with 4.5.3, the antenna assembly consisting of the RF amplifier, housing assembly, mating connectors, and cable sections, specified in items b, c, d, e, and f of FIGURE 1, shall exhibit no physical or electrical degradation as revealed by surface examination and pressure test. The cable termination tip, specified in item a of FIGURE 1, shall exhibit no physical or electrical damage as revealed by surface examination. The complete antenna assembly shall conform to all other requirements specified herein.

3.2.3.4 Durability. The complete antenna assembly with the exception of the inboard connector, UG-1820/U, shall withstand the durability test of 4.5.4. The antenna assembly shall exhibit no degradation when subjected to the quality conformance inspection specified in 4.4 and tests specified in 4.5, except the test specified in 4.5.3.

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3.2.3.5 Pressure resistance. When tested with hydrostatic pressure in accordance with 4.5.10, the RF amplifier shall show no damage or evidence of flooding. If either connector is opened subsequent to the pressure test and before delivery, the test shall be repeated after reassembly.

3.3 Design and construction. The antenna assembly configuration shall be in accordance with NUSC/NLL D 03056001. The RF amplifier schematic shown on NUSC/NLL D 03055004 may be used as a guide to achieve the electrical performance requirements specified in 3.3.1.

3.3.1 Electrical design. Polarity and grounding of circuits shall ensure compatibility with couplers and minimize adverse electrolysis. The RF amplifier shall be designed to operate from a constant voltage source of 6 volts direct current (VDC) connected with the positive polarity on the center conductor and the negative polarity on the outer conductor of the buoyant coaxial RF cable assembly. The current drawn by the RF amplifier shall be 26 ±3 mA. The output of the RF amplifier shall be connected between the center conductor and the outer conductor of the RF cable assembly. The outer conductor of the RF cable assembly shall be grounded to sea water at the connector, plug, electrical (socket contact) as shown on NUSC/NLL D 02382001.

3.3.2 Materials and parts. Materials and parts shall be selected in accordance with MIL-E-16400. Materials and parts shown on the DWGs are approved for the application shown without further reference to the Government.

3.3.2.1 RF amplifier construction. The RF amplifier chassis design shown on NUSC/NLL D 03055001 is provided as guidance and may be reconfigured to support the contractor's design, providing the design conforms to the requirements of this specification and specifically the flexibility and pressure tests.

3.3.2.2 Nondeviation parts. The components specified in a through f are considered nondeviation with respect to design and construction and shall be manufactured and assembled in accordance with the DWGs and specifications specified in a through f:

	Component	Document
a.	Tip, Cable Termination (Grounded)	NUSC/NLL 02202001
b.	Cable, RF Single Conductor, Type I	NUSC-C-342/4141-279/1-10
c.	Cable, RF Coaxial, Type II	NUSC-3-342/4140-279/2-12
d.	Connector, Plug, Electrical (Pin Contact)	NUSC/NLL D 02379001
e.	Connector, Plug, Electrical (Socket Contact)	NUSC/NLL D 02382001
f.	Housing Assembly	NUSC/NLL D 02380001

3.4 Marking. The antenna assembly shall conform to the marking and identification letter requirements of DWGs, parts lists, specifications, and other documents shown on NUSC/NLL D 03056001.

3.5 Workmanship. The antenna assembly shall be manufactured with care using the exact materials (see 3.3.2), dimensions (see 3.2.2.1), and surface finish (see 3.2.2.1). Assembly shall ensure correct arrangement, hookup, and fastening of all components. The molding procedures for the connectors and the housing are particularly critical and shall be performed with careful attention to time, temperature, spacing and lay of wires, and of housing reinforcement end tension. The completed unit shall be clean and free of burrs, cuts, gouges, and grit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.3 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections. Testing shall be performed at ambient pressure and room temperature unless otherwise specified in the respective test paragraphs of 4.5.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (see 4.4.1)
c. Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. The number of units to be submitted for first article inspection shall be as specified in the contract (see 6.2.1). First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III. Order of testing shall be as listed in TABLE III.

TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection (production)
Burn-in	3.2.1.1	4.5.1	X	X
Tensile strength	3.2.3.2	4.5.2	X	
Flexibility	3.2.3.3	4.5.3	X	X
Durability	3.2.3.4	4.5.4	X	
Marking	3.4	4.5.5	X	
Voltage gain	3.2.1.2	4.5.6	X	X
Stability	3.2.1.3	4.5.7	X	
Noise figure	3.2.1.4	4.5.8	X	
Gain compression	3.2.1.5	4.5.9	X	
Pressure resistance	3.2.3.5	4.5.10	X	X
Physical characteristics	3.2.2	4.5.11	X	X
Nondeviation parts	3.3.2.2	4.5.11	X	
Materials and parts	3.3.2	4.5.11	X	
Workmanship	3.5	4.5.11	X	X

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 and 4.4.2.

4.4.1 Production inspection. Production inspection shall be conducted on each antenna assembly offered for delivery. Production inspection shall comprise such examination and testing which will prove the workmanship and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, and tests which detect hidden defects of materials. Quality conformance inspection shall include the tests specified in TABLE III.

4.4.2 Nonconforming production units. If more than three units per 50 units tested fail the production inspection specified in 4.4.1, the contractor shall immediately investigate the cause of failure to determine if the redesign permitted in 3.3 has affected reliability. Any reject rate associated with a specified component, process, or assembly in excess of that specified shall require testing, as specified for the first article, of an additional acceptance item from the same lot.

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4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.11.

4.5.1 Burn-in test. The amplifier shall be subjected to a 24-hour test with a constant 6.5 VDC supplied through the coupling network (see FIGURE 2).

4.5.2 Tensile test. The tensile strength of the connectors and housing shall be demonstrated by subjecting a test assembly to a static load of 300 pounds for 4 hours at room temperature. The length of the assembly shall be measured under load at the beginning and end of the test and the sample examined for evidence of cracking, stretching, or other damage.

4.5.3 Flexibility test. The molded termination tip, RF amplifier, and mating connectors shall be flexed through an S-bend path in a test device similar to FIGURE 3, three times in each direction, at any convenient rate. The test shall be conducted at room temperature. The components shall subsequently pass all the other tests and inspections in TABLE III.

4.5.4 Durability. Compatibility with the Antenna Transfer Assembly, AN/BRA-24( ) shall be demonstrated by subjecting the antenna assembly to 25 cycles, out and in, through the Antenna Transfer Assembly, AN/BRA-24( ), pressurized to 100 ±10 pounds per square in. gage.

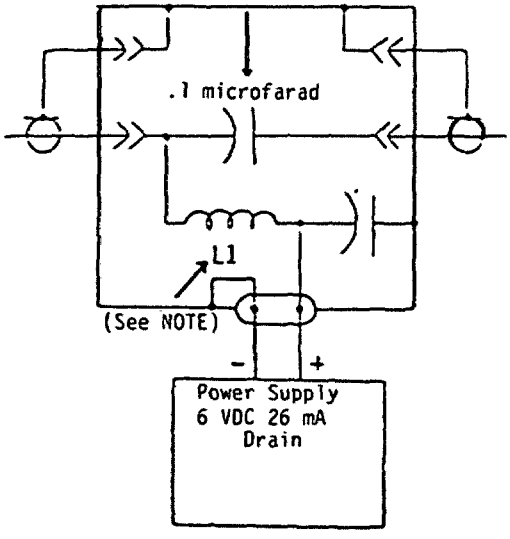
4.5.5 Marking examination. The marking examination shall consist of verifying that the position marks and identification letters located along the length of the antenna assembly, as shown in FIGURE 4, are legible at a distance of 3 ft and that they are an integral part of the antenna assembly.

4.5.6 Voltage gain test. The RF amplifier (see NUSC/NLL D 03055001), assembled in the housing (see NUSC/NLL D 02380001) with test adapters (see NUSC/NLL D 02383001 and D 02384001) at each end shall be tested for voltage gain as specified in a through f:

a. Connect the amplifier in the test setup shown in FIGURE 5.
b. Set the step attenuator to 40 dB and the tracking generator output to minimum (-10 dBm) to avoid saturation of the amplifier.
c. Shut off the power supply and remove the amplifier from the circuit, replacing the amplifier with a 50-ohm cable. Select the frequency range to be displayed.
d. Set a convenient reference level near the bottom of the screen.
e. Replace the amplifier, turn on the power supply, and use a camera or plotter to record the gain versus frequency curve.
f. Repeat c, d, and e, as necessary, to cover the range specified in TABLE I.

4.5.7 Stability test. The RF amplifier, assembled in the housing with test adapters (see NUSC/NLL D 02383001 and D 02384001) at each end, shall be tested for stability as specified in a and b:

a. Connect the amplifier in the test setup shown in FIGURE 6 with the power supply activated.
b. Scan the 0.3-MHz to 1250-MHz spectrum to detect any signals that may be present. No signals shall be observed.



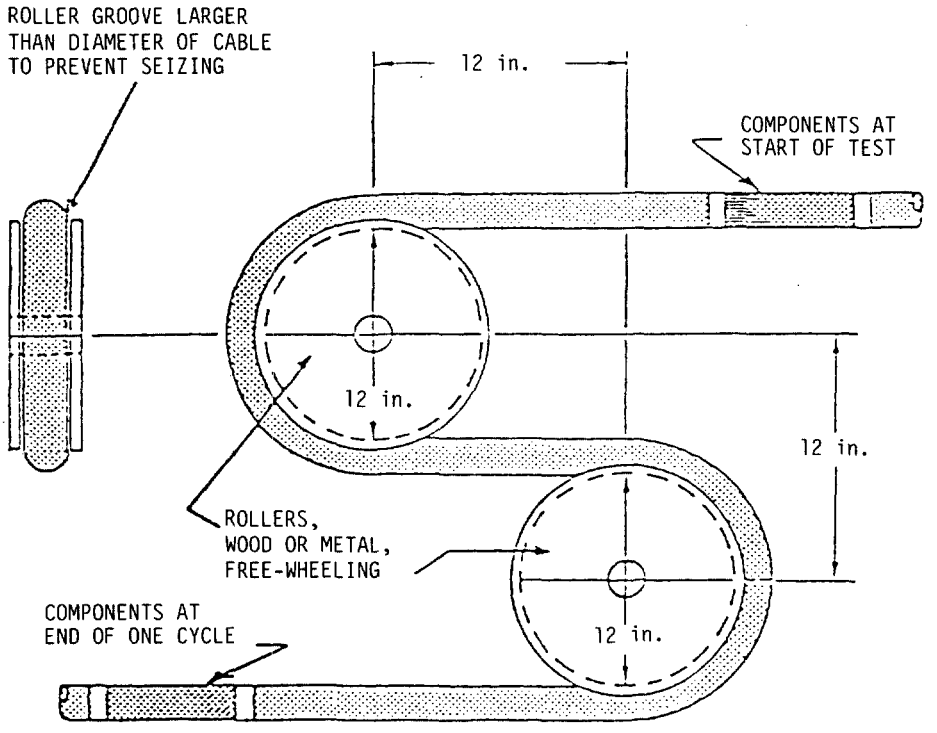
Coupling network to be built-up in Pomona Electronics Company Shielded Box Model 2417, or equal.

Coupling network to be built-up in Pomona Electronics Company Shielded Box Model 2417, or equal

NOTE: L1 = 120 microhenry RF choke (Lenox-Fugle NR-120, or equal)

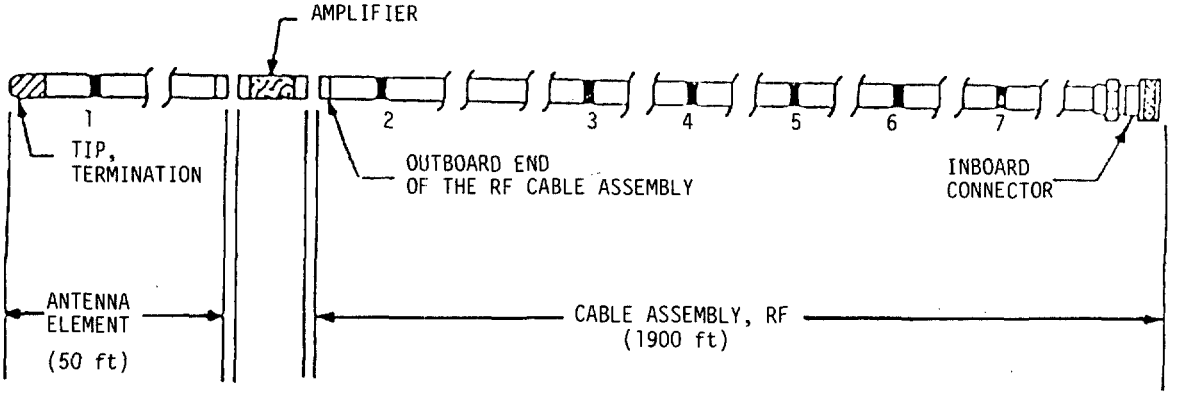
FIGURE 2. Coupling network.

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NOTES:
1. Use tension enough to bend antenna assembly into close contact with rollers, ensure that the molded termination tip, housing assembly, and mating connectors are held against rollers, making contact through at least 180 degrees of each roller.
2. Cable can be the complete antenna assembly or test sections sufficient to allow components to be drawn around rollers.

FIGURE 3. Flexibility test device.



Mark No.

1 Located on the antenna element: 30 ft +1 ft from termination tip to identify when the tip has passed through the AN/BRA-24 and all associated shear, shut-off, ball, and back valves.

2 Located on the RF cable assembly: 30 ft +1 ft in from the outboard end of the RF cable assembly to identify when the amplifier is in a protected environment within the conduit or approaching the AN/BRA-24.

3-7 Located on the RF cable assembly: 1000 ft, 1200 ft, 1400 ft, 1600 ft, and 1800 ft ± 5 ft in from the outboard end of the RF cable assembly.

FIGURE 4. Position mark location.

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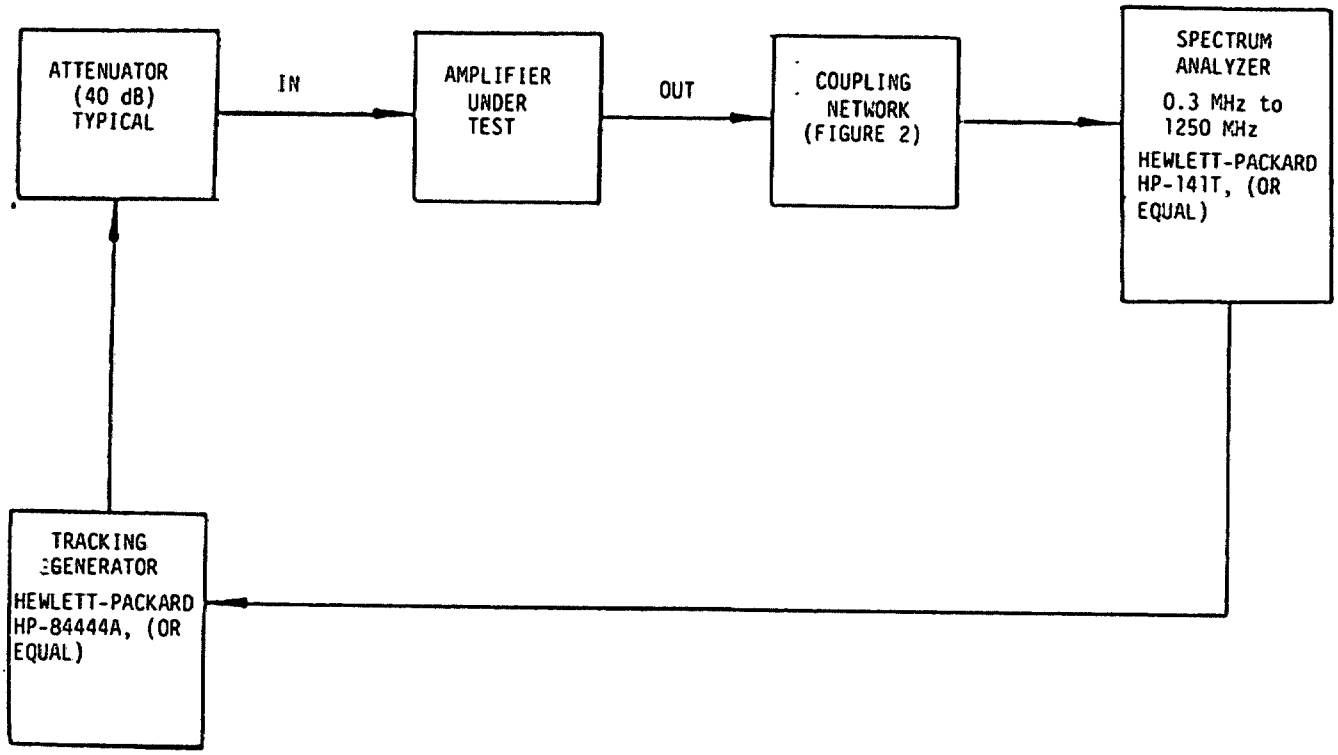


FIGURE 5. Voltage gain test setup.

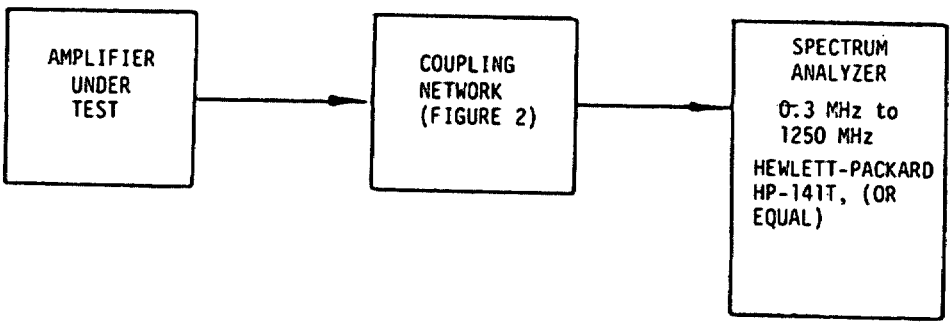


FIGURE 6. Stability test setup.

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4.5.8 Noise figure test. The RF amplifier, assembled in the housing with test adapters (see NUSC/NLL D 02383001 and NUSC/NLL D 02384001) at each end shall be tested for noise figure as specified in a through d:

a. Connect the amplifier in the test setup shown in FIGURE 7. A setup using an automatic noise figure meter may be substituted if available. Activate the amplifier under test.
b. Turn the noise source to minimum and set the receiver to a convenient reference level at 2 MHz.
c. Increase the output of the noise source with the received noise 3 dB above the reference level, and record the noise figure indicated on the noise source.
d. Repeat b and c at 10 MHz and 30 MHz

4.5.9 Gain compression test. The RF amplifier, assembled in the housing with test adapters (see NUSC/NLL D 02383001 and NUSC NLL D 02384001) at each end shall be tested for gain compression as specified in a through e:

a. Connect the amplifier in the test setup shown in FIGURE 8 with the VLF/VHF coupler activated and the signal generator and spectrum analyzer set at 2 MHz.
b. Adjust signal generator amplitude to produce -30 dBm in the spectrum analyzer display with the external step attenuators set at zero dB.
c. Increase the signal generator amplitude by 10 dBm and switch in 10 dB of loss with the step attenuators. Continue increasing the level and compensating with attenuators in 10 dBm and then 1 dBm steps until the level in the spectrum analyzer display has dropped to -31 dBm.
d. Determine amplifier output level for 1 dB of gain compression as follows and verify compliance with the requirements of 3.2.1.5:

Output level (dBm) = -31 plus the setting of 1 dB step attenuator plus the setting of 10 dB step attenuator.
e. Repeat b and d with the signal generator and spectrum analyzer set to 10 MHz and 30 MHz.

4.5.10 Pressure resistance test. The pressure resistance test shall be performed as specified in 4.5.10.1 through 4.5.10.4.

4.5.10.1 Antenna assembly. The complete antenna assembly shall be tested for watertight integrity as specified in a through d:

a. Subject the RF amplifier and connectors that mate with the antenna element assembly and the RF cable assembly to a hydrostatic pressure of 400 pounds per square in. (psi) for 2 hours. The conductivity of the water shall be at least 0.1 mho per meter. One level teaspoon of baking soda per gallon of water conforms to this requirement. The pressure vessel can be doubled-ended and just long enough to contain items c, d, e, and a portion of b and f of FIGURE 1.
b. Remove the test assembly from the pressure vessel and dry water off the surface.
c. Using a Simpson 260 volt-ohmmeter or equal (a digital voltmeter (DVM) will not work as silicon diodes are being measured), measure the resistance between the grounding tip of the antenna element and the body of the RF amplifier. The resistance shall be 14 ohms +5 ohms for either polarity.
d. Using DVM or Simpson 260, or equal, measure the resistance between the center conductor and shell of the UG-1820/U connector. The resistance shall measure 1260 ohms ±100 ohms.

4.5.10.2 Antenna Element Assembly, AS-3376/BRR. When delivered as a separate item, the antenna element assembly shall be tested for watertight integrity as specified in a through e:

a. Install a test plug made in accordance with NUSC/NLL C 02381001 in the unmated end of the RF amplifier.
b. Subject the RF amplifier and connectors that mate with the antenna element assembly and test plug to a hydrostatic pressure of 400 psi for 2 hours. The conductivity of the water shall be at least 0.1 mho per meter. One level teaspoon of baking soda per gallon of water conforms to this requirement.
c. Remove the test assembly from the pressure vessel and dry water off the surface.
d. Using a Simpson 260 DVM, or equal (a DVM will not work as silicon diodes are being measured), measure the resistance between the grounding tip of the antenna element and the body of the amplifier. The resistance shall be 14 ohms ±5 ohms for either polarity.

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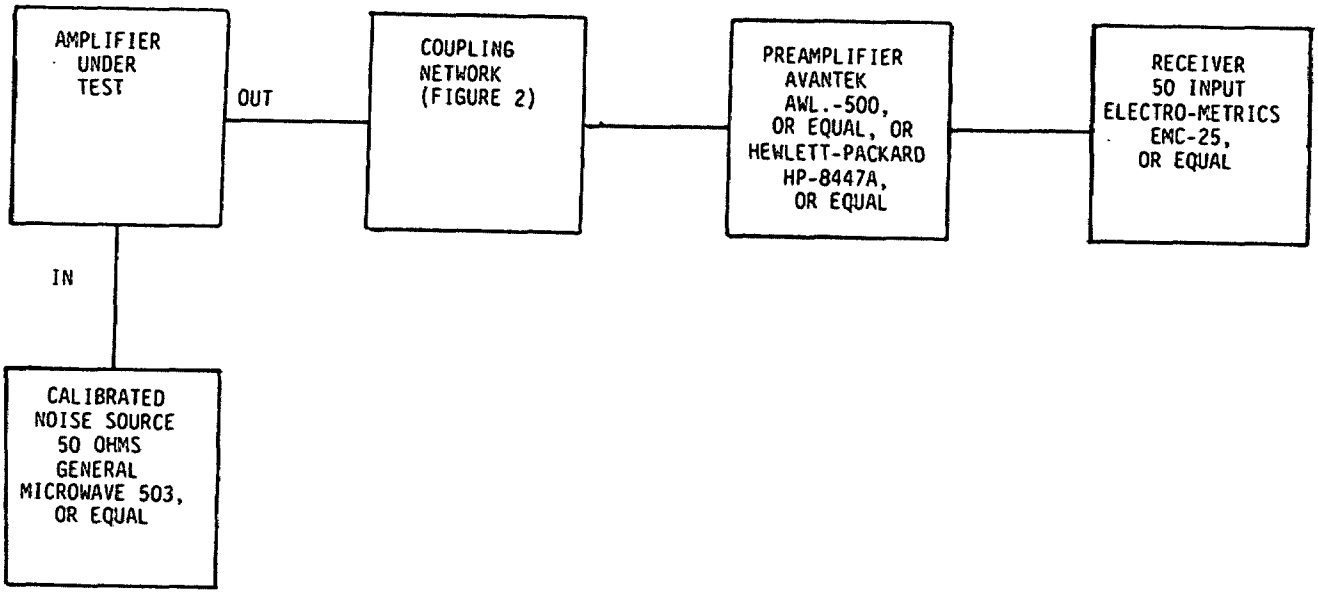


FIGURE 7. Noise figure test setup.

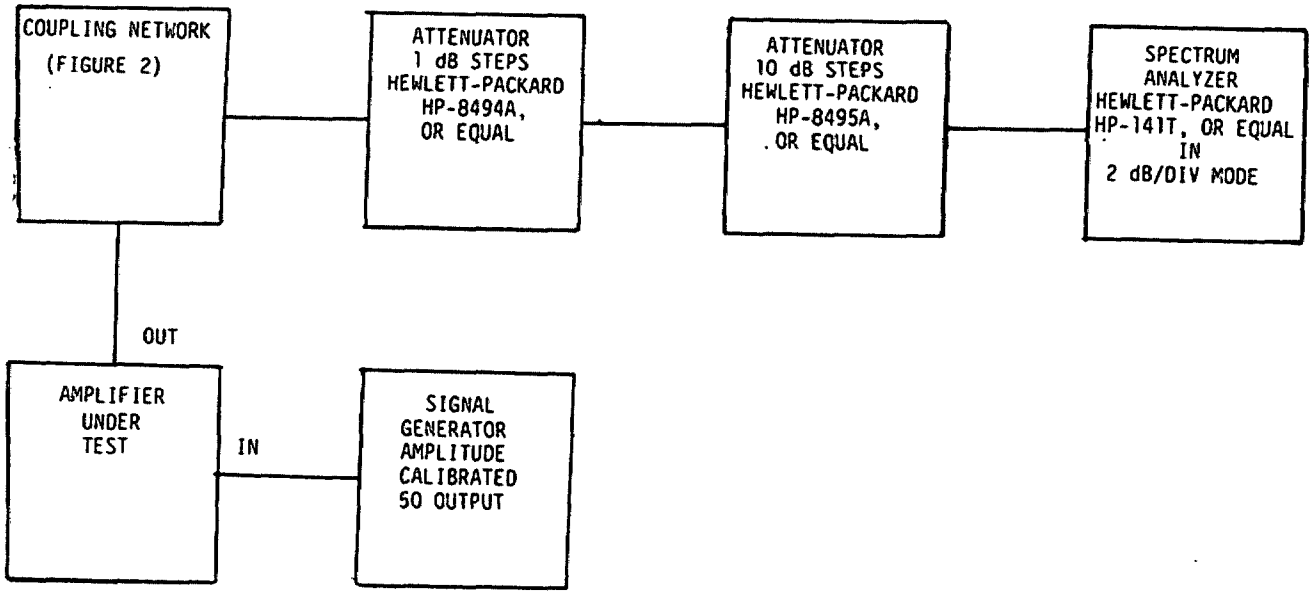


FIGURE 8. Gain compression test setup.

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e. Remove the test plug and inspect the RF amplifier for water leakage. If dry, reinstall the test plug to protect the amplifier during shipment. The presence of moisture constitutes failure of this test.

4.5.10.3 Cable Assembly, RF CG-3816/BRR. When delivered as a separate item, the RF cable assembly shall be tested for watertight integrity as specified in a through c:

a. Place a cover assembly made in accordance with NUSC/NLL C 02377003 over the connector, plug, electrical (socket contact) shown on NUSC/NLL D 02382001.

b. Subject the connector (socket contact) and cover assembly to a hydrostatic pressure of 400 psi for 2 hours. The isolation resistance between the center conductor and body of the Connector, Plug, Electrical, UG-1820/U, shall be at least 1000 megohms during the hydrostatic test. The conductivity of the water shall be at least 0.1 mho per meter. One level teaspoon of baking soda per gallon of water conforms to this requirement.

c. Remove the test assembly from the pressure vessel and dry water off the surface. Leave the cover assembly in place to protect the connector (socket contact) during shipment (see 5.4). Removal of the cover assembly will void the results and the test shall be repeated.

4.5.10.4 Operating tests. Satisfactory operation after environmental testing shall be demonstrated by performing the voltage gain test of 4.5.6 at 1 MHz and 30 MHz only.

4.5.11 Surface examination. Surface examination shall be performed in accordance with MIL-E-16400. The requirements of 3.2.2.1, 3.2.2.2, 3.3.2, and 3.5 shall be verified during this examination.

4.6 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Antenna assembly. No preservative shall be applied to the antenna assemblies. Packaging shall be on nonreturnable reels in accordance with Level A of MIL-C-12000, except that the wood lagging shall be nominal 1-in. thick and shall be strapped with three 0.625-in. by 0.030-in. nonmetallic straps conforming to PPP-S-760, Type II. Packaging shall afford protection against physical damage during shipment, handling, uncovered storage, and worldwide distribution. The reel shall contain a suitable slotted partition so the antenna element assembly can be wound separately from the RF cable assembly. The cable termination tip and RF amplifier assembly shall be accessible for checking continuity. The RF amplifier assembly shall be wrapped with cushioning material (Acrotube or Armoflex pipe insulation, or equal) for extra protection. The Connector, Plug, Electrical, UG-1820/U, shall be at the circumference of the reel and protected with a cap, MX-1143A/U (see NUSC/NLL D 03056002). Reels shall have a hub diameter not less than 18 in., a width not more than 26 in., and an outside diameter not more than 36 in.

5.2 Antenna Element Assembly, AS-3376/BRR. When delivered as a separate item, the antenna element assembly shall be shipped in a coil not over 22 in. in diameter, packaged in accordance with Level A of MIL-C-12000, except that the shipping container shall conform to PPP-B-640, Class 2, Style E and strapped with two 0.50-in. by 0.020-in. nonmetallic straps conforming to PPP-S-760, Type II, to afford protection against physical damage during shipment, handling, uncovered storage, and worldwide distribution. The unmated end of the RF amplifier shall be protected with the test plug (see NUSC/NLL C 02381001). The RF amplifier shall be wrapped for extra protection as specified in 5.1. The box containing the antenna element shall be small enough to fit through a 25 in. diameter hatch, but the inside diameter of the cable in the coil shall be 15 in. or more. An instruction sheet for the connector assembly and screws for assembling the antenna element assembly to the RF cable assembly shall be packed in each box.

5.3 RF Amplifier, AM-7041/BRR. When delivered as a separate item, the RF amplifier shall be disassembled into three components: RF amplifier (see NUSC/NLL D 03055001), housing (see NUSC/NLL D 02380001), and liner (see NUSC/NLL D 02380005). Gain test results made in accordance with 4.5.6, a copy of the assembly instructions (see NUSC/NLL A 02710001), and screws for assembling the RF amplifier to the antenna element assembly and the RF cable assembly shall be packaged with each RF amplifier in a plastic bag. The RF amplifier and housing shall be packaged in a box, conforming to V3c of PPP-B-636 with additional packing to afford protection against physical damage during shipment, handling, and covered storage. The top and bottom flaps of the box shall be sealed by the application of 2-in. wide tape conforming to PPP-T-60, Type IV. The box dimensions shall not exceed those specified in a through c:

a. Length: 12 in.
b. Width: 4 in.
c. Depth: 4 in.

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5.4 Cable Assembly, RF, CG-3816/BRR. No preservative shall be applied to the cable assembly. Packaging shall be on nonreturnable reels in accordance with Level A of MIL-C-12000, except wood lagging shall be nominal 1-in. thick and shall be strapped with three 0.625-in. by 0.030-in. nonmetallic straps conforming to PPP-S-760, Type II. Packaging shall afford protection against physical damage during shipment, handling, uncovered storage, and worldwide distribution. The connector, plug, electrical (socket contact), (see NUSC/NLL D 02382001) shall be accessible for checking continuity. The connector (socket contact), cover assembly, and the Connector, Plug, Electrical, UG-1820/U shall be wrapped with cushioning material (Acrotube or Armoflex pipe insulation, or equal) for extra protection. The Connector, Plug, Electrical, UG-1820/U, shall be at the circumference of the reel and protected with a cap, MX-1143A/U (see NUSC/NLL D 03056002). Reels shall have a hub diameter not less than 18 in., a width not more than 26 in., and an outside diameter not more than 36 in.

6. NOTES

6.1 Intended use. The antenna assembly covered by this specification is intended for use by submarines.

6.2 Ordering data. Ordering data should be as specified in 6.2.1 and 6.2.2.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification
b. Length of antenna assembly if other than specified in 3.2.2.1
c. Number of first article samples to be submitted (see 4.3)
d. Levels of preservation, packaging, packing, and marking (see 5.1 through 5.4)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of the Defense Acquisition Regulation 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.3	First Article Inspection Procedure	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902
4.4	Test Plan (Nonsystem)	DI-T-3709
4.5	Environmental Procedures Test	UDI-T-22710B

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it should be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article should be a first production item. The first article should consist of the number of antenna assemblies specified in the contract. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project No. 5985-N519)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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